Electrochemistry
The Electrochemical Society of Japan
Received: April 15, 2013
Accepted: July 10, 2013
Published: October 5, 2013
http://dx.doi.org/10.5796/electrochemistry.81.808
JOI:DN/JST.JSTAGE/electrochemistry/81.808
Communication
Electrochemistry, **81(10)**, 808–810 (2013)

Molecular Dynamics Study of Ionic Liquids in Graphite Nanopores

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ABSTRACT
We performed MD calculations on ionic liquids (ILs) to investigate the effects of solvent and pore size. The slit graphite pore models with three pore sizes 1.6, 2.8, and 4.0 nm were constructed, which confine EMITFSI and EMITFSI/PC. We found that EMI^+ and solvent PC form layering structures along the pore walls. For the effects of solvent, the layering of EMI^+ is interfered by PC. It was also clarified, for the effects of pore size, that the diffusion coefficients of EMI^+ and $TFSI^-$ in 1.6 nm pore are significantly large compared with other pore sizes. In the 1.6 nm pore, EMI^+ forms triple layering structures. Since this pore size nearly corresponds to the threefold space of molecular thickness, we concluded that the layering of ILs is important for the ionic mobility improvement in nanopores.

Keywords : Electrical Double Layer, Ionic Liquids, Molecular Dynamics

1. Introduction

Electrical Double Layer Capacitors (EDLCs) using ionic liquids (ILs) have been expected as new power sources for their high power density and high thermal stability.[1,2] The electric properties of EDLCs are mainly determined by the mobility of ILs and the pore size of activated porous graphite electrodes. There are many studies on the effects of graphite pore size.[3–7] For the ILs; 1-ethyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide (or amide) (EMITFSI) system, Largeot et al.[3] reported the existence of capacitance peak near 0.7 nm pore size, experimentally. Later, Feng and Cummings[4] found another capacitance peak around 1.4 nm pore size by molecular dynamics (MD) calculations, and suggested oscillatory behavior of capacitance as a function of pore size. Rajput et al.[5] examined the effects of slit pore size and pore loading on the properties of ILs by MD calculations. However, the relationship between oscillatory behavior of capacitance and dynamics of ILs has not properly been explained. Furthermore, the effects of solvent in nanopores are hardly understood. In the present study, we performed the classical MD calculations to investigate the effects of solvent (propylene carbonate: PC) and graphite pore size.

2. Method

The full atomistic MD calculations were performed by using VSOP program in J-OCTA[8] package. In Fig. 1, we show our slit pore structural model, which consist of graphite planes and ILs. Two kinds of ILs system, EMITFSI and EMITFSI/PC, were examined to analyze the effects of solvent PC. The molecular structures are shown in Fig. 2. Then, three pore sizes of 1.6, 2.8, and 4.0 nm were chosen to analyze the effects of graphite pore size. The densities of ILs were settled to be similar to bulk MD results, i.e., 1.62 g/cm^3 for bulk EMITFSI, and 1.39 g/cm^3 for bulk EMITFSI/PC. Carbon atoms of graphite planes were modeled by the Lennard-Jones particle with $\sigma_c = 0.340$ nm and $\varepsilon/k_B = 43.3$ K (AMBER param99), and they were fixed on MD calculations. We considered the electrically neutral condition, i.e., the graphite planes were not electrically charged, and the number of EMI^+ and $TFSI^-$ were equal. The content ratio of ILs in EMITFSI/PC system was settled at 48.9 wt%. The force field parameters were taken from Liu et al.[9] for EMI^+, Lopes and Padua[10] for $TFSI^-$, and Takeuchi et al.[11] for PC. Atomic charges of molecules were obtained by the ab initio restrained electrostatic potential fitting (Gaussian B3LYP/6-311+G(d) level calculations). All MD calculations were carried out in the canonical (NVT) ensemble with the Nose-Hoover thermostat. The cutoff distance of the Lennard-Jones interactions

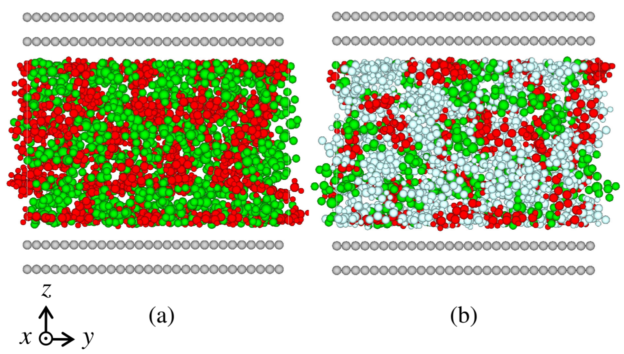

Figure 1. (Color online) Snapshots of slit graphite pore models (2.8 nm), which confines (a) EMITFSI and (b) EMITFSI/PC. Atoms of EMI^+, $TFSI^-$, PC, and graphite are depicted in red, green, light blue and gray balls, respectively.

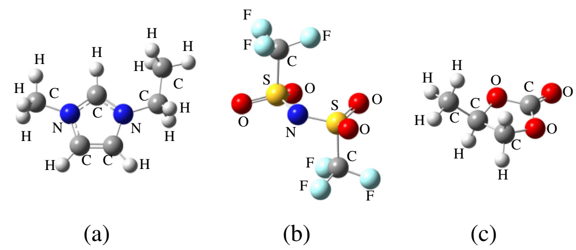

Figure 2. (Color online) Schematic representation of electrolyte molecules (a) EMI^+ (b) $TFSI^-$ and (c) PC.

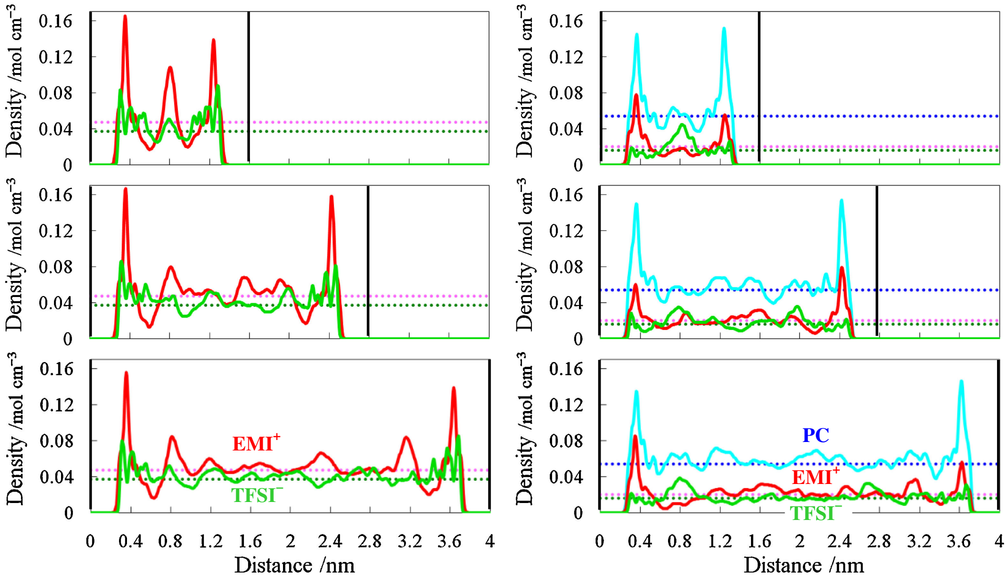

Figure 3. (Color online) Number density profiles along *z*-direction. These were calculated from the summation of every atom which forms the each molecule (EMI$^+$, TFSI$^-$, and PC). The horizontal axis denotes the distance from one side of graphite wall. The vertical black bold lines represent the positions of graphite wall at 0.0, 1.6, 2.8, and 4.0 nm. The dotted lines represent the averaged bulk model results.

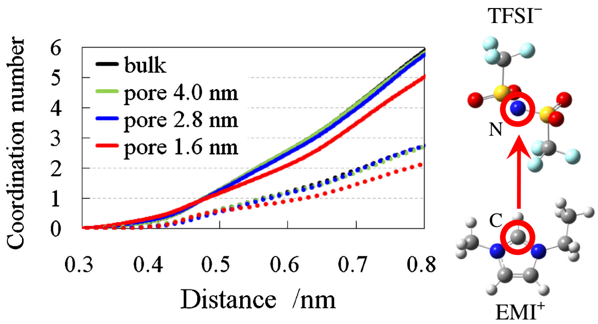

Figure 4. (Color online) Coordination number of TFSI$^-$ around EMI$^+$. The horizontal axis represents the distance between the carbon atom of EMI$^+$ and the nitrogen atom of TFSI$^-$, as shown here. Solid and dotted lines show the results of EMITFSI and EMITFSI/PC system, respectively.

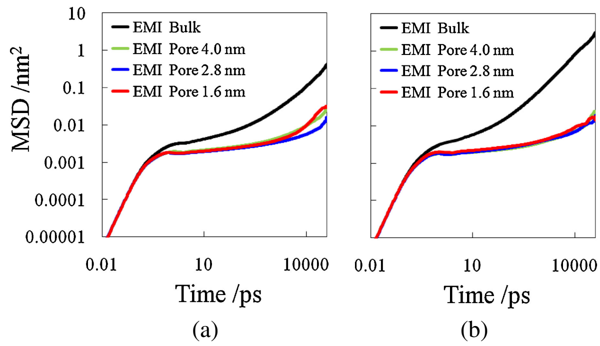

Figure 5. (Color online) MSD of EMI$^+$ along the inplane *xy*-direction confined in bulk and in graphite pore. (a) is the results of EMITFSI and (b) is EMITFSI/PC.

was 1.0 nm, and the long range coulomb interactions were handled by the particle-mesh Ewald (PME) method with the cutoff distance of 1.2 nm and grid intervals were 0.11 nm in the (x,y) directions and 0.31 nm in the *z*-direction. The three dimensional periodic boundary conditions were applied. All molecules were located in the center of orthorhombic cell ($3.57 \times 3.53 \times 10.0\,\mathrm{nm}^3$), with vacuum layer (larger than 5.2 nm) on each side of slit graphite pore.

In order to obtain the thermally equilibrium initial structures, we performed preparation MD calculations as follows. First, we optimized the slit pore size. Second, MD systems were melted at 600 K for 1 ns. Third, they were annealed in 1 ns at 500 K, in 1 ns at 400 K, and in 2 ns at 300 K. At last, after preparation calculations, we executed 40 ns MD calculations, with a time step of 2 fs.

3. Results and Discussion

Firstly, we explain the effects of solvent PC on structures and dynamics. In Fig. 3, we show the number density profiles along the *z*-direction. Planar shaped molecules, EMI$^+$ and PC, have large density peaks near the pore-walls (0.0, 1.6, 2.8, and 4.0 nm). That is, they are aligned along the graphite pore-walls. In particular, in the 1.6 nm pore, EMI$^+$ forms triple layering structure. However, this layering structure is interfered by PC. This comes from the molecular shape difference between EMI$^+$ and PC, i.e., PC is quasi-planer molecule which have methyl group. In Fig. 4, we show the coordination number of TFSI$^-$ around EMI$^+$ as a function of the distance between them. We clearly see that coordination numbers are decreased by the solvation effect of PC. Thus, the coulombic interactions between EMI$^+$ and TFSI$^-$ are reduced by PC.

In Fig. 5, we show mean square displacement (MSD) results of EMI$^+$ along the inplane *xy*-direction. Three typical time regions were observed in MSD, the ballistic region (up to 0.5 ps), the sub-diffusive/intermediate region and the diffusive/Fickian region (over 10 ns), respectively. In the ballistic region, log[MSD] is proportional to the twice of log[time]. Thus, the molecules can move as linear uniform motion. On the other hand, in the diffusive region, log[MSD] shows the linear increase as a function of log[time]. This is the results of interaction with other molecules. In Table 1, we show the results of self diffusion coefficient at 300 K, which were calculated by the Einstein's equation with the gradients of MSD in the diffusive time region. The MSD component of the inplane *xy*-direction is larger than that of *z*-direction, so the data has not been shown in this figure. In the bulk model, the diffusion coefficients of EMI$^+$ and TFSI$^-$ were increased nearly eight times by the solvation of PC. This is the effect of the decrease in coordination number of

Table 1. Diffusion coefficients along the inplane *xy*-direction.

System	Molecule	Diffusion coefficient/$10^{-12}\,m^2\,s^{-1}$			
		Slit graphite pore size			Bulk
		1.6 nm	2.8 nm	4.0 nm	
IL	EMI^+	0.24	0.09	0.16	2.5
	$TFSI^-$	0.20	0.08	0.12	1.5
IL with solvent	EMI^+	0.14	0.10	0.16	20.2
	$TFSI^-$	0.18	0.09	0.15	12.5
	PC	0.22	0.16	0.21	31.5

$TFSI^-$ around EMI^+. However, the solvation effect of PC on the ILs mobility is drastically diminished in graphite nanopore.

Secondly, we also present the effects of graphite pore size on structures and dynamics. As we see in Fig. 3, the density profiles in far region from pore walls are getting closer to the averaged bulk model results with increasing the pore size. On the other hand, the density profiles in near region to pore walls hardly depend on the pore size, i.e., density profiles in the near region simply depend on the distance from graphite. This result reveals the spontaneous formation of electrical double layer near the pore walls. Namely, the influence from the near region decrease with increasing the pore size. In the near region to pore walls, electrolyte molecules are condensed by the interaction with the graphite pore walls, i.e., the mobility of ILs and solvent are reduced. We consider this is the cause of the pore size effects on structures and dynamics. As mentioned above, in EMITFSI system with the 1.6 nm pore, EMI^+ forms the triple layering structure. On the other hand, in EMITFSI/PC system, the triple layering structure of EMI^+ is interfered by PC, i.e., the central peak of EMI^+ is diminished. However, the triple layering structure is also formed in EMITFSI/PC system, with the combination of EMI^+, $TFSI^-$, and PC. We propose that these layering structure formations depend on molecular sizes. Since this pore size nearly corresponds to the threefold space of EMI^+ and PC molecular thickness. Thus, if the other ILs systems are examined, this pore size will change according to the ILs sizes. Moreover, as shown in Fig. 4, this layering structure results in the decrease of coordination number between EMI^+ and $TFSI^-$, comparing to other pore sizes and bulk results. Namely, the coulombic interactions between EMI^+ and $TFSI^-$ were reduced by the layering effect. In other words, the electrical double layer structure was clearly formed in 1.6 nm pore. We consider this electrical double layer structural change is the cause of the capacitance increase of EDLCs, as reported by Feng and Cummings.[4] Because our MD results were obtained under the condition of electrically neutral graphite, the electrical double layer structure in 1.6 nm pore is to be affected under the presence of electric field. However, we expect the EDLCs capacitance to be enhanced by the layering effect in particular pore size. Furthermore, the diffusion coefficients of EMI^+ and $TFSI^-$ in 1.6 nm pore are significantly large compared with those of other pore sizes, as shown in Table 1. This is the new finding of the present study. Rajput et al.[5] reported that the diffusion coefficients of EMI^+ and $TFSI^-$ show the monotonically decrease behavior with decreasing pore size. They examined the pore size over 1.9 nm. However, in the present study, the diffusion coefficients of ILs turn to increase by the layering effects in the smaller pore size under 1.9 nm. Namely, according to the formation of the triple layering structure, the resistances on dynamics, i.e., the molecular interactions between three molecular layers were reduced. This will lead to the increase of ILs mobility in 1.6 nm pore.

4. Conclusion

MD calculations were performed to study the properties of ILs confined in slit-like graphite pore. We found that EMI^+ and solvent PC form layering structures along the pore walls. For the effects of solvent, the layering of EMI^+ is interfered by PC. Then, in the bulk model, the diffusion coefficients of ILs are increased by PC. However, the solvation effect of PC on the ILs mobility is drastically diminished in graphite nanopore. It was also clarified, for the effects of pore size, that the diffusion coefficients in the 1.6 nm pore are significantly large compared with other pore sizes. In the 1.6 nm pore, EMI^+ forms triple layering structures, since this pore size nearly corresponds to the threefold space of molecular thickness. Namely, according to the formation of the triple layering structure, the resistances on dynamics, i.e., the molecular interactions between three molecular layers were reduced. This will lead to the increase of ILs mobility in 1.6 nm pore. Thus, we concluded that the layering of ILs plays important role for their mobility and formation of the electrical double layer in nanopores.

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